SUPPORTING ADULT GUIDANCE

The angle in a semicircle

Circle Theorems · Lesson 3. For any supporting adult — teacher, practitioner, parent or carer.
You do not need to be a mathematics specialist to support this lesson.

What this lesson does

This lesson establishes that an angle inscribed in a semicircle (subtended by a diameter) is a right angle — Thales' theorem — as the first corollary of the angle-at-the-centre theorem. Learners discover the invariant right angle by dragging a point in the Circle Theorem Lab (diameter case), then prove it in three lines from the keystone, and meet the useful converse. The result is then used, with reasons, to find angles in right-angled triangles inscribed in circles.

Driving Question: *What is special about an angle drawn on a diameter?*

The mathematics, in plain terms

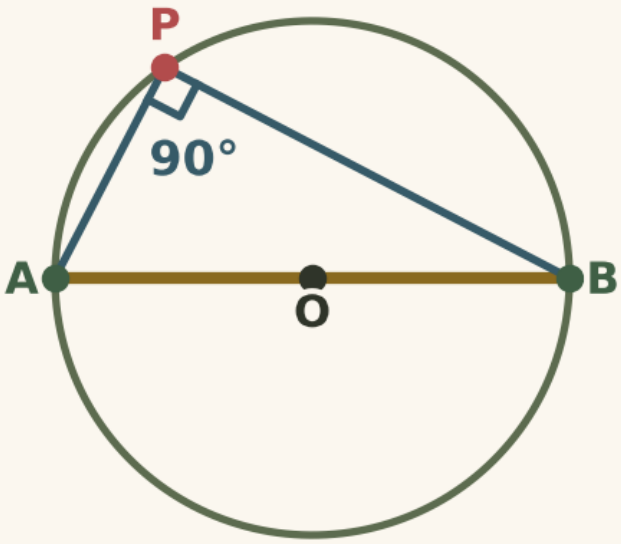
If AB is a diameter of a circle and P is any other point on the circle, then ∠APB = 90°. Proof: the diameter subtends a straight angle (180°) at the centre, and by the angle-at-the-centre theorem the inscribed angle is half of it, i.e. 90°. Equivalently, joining OP creates two isosceles triangles with base angles a and b, and since 2a + 2b = 180° the angle at P (a + b) is 90°. The converse holds: the hypotenuse of a right-angled triangle is a diameter of its circumscribed circle.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Find the missing angle

AB is a diameter and P is on the circle. One acute angle, ∠PAB, is 35°. Find ∠PBA.

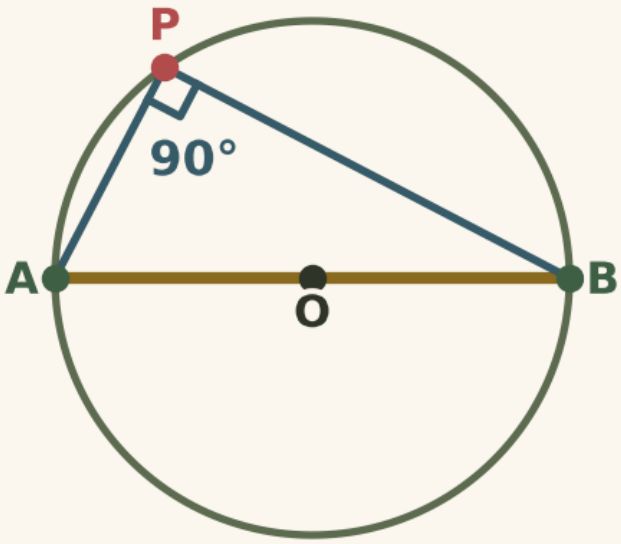


1. AB is a diameter, so ∠APB = 90°.	angle in a semicircle
2. Angles in triangle APB sum to 180°.	35 + 90 + ∠PBA = 180
3. So the two acute angles add to 90°.	∠PBA = 90 – 35

∠PBA = 55°, because the angle in a semicircle is 90° and the angles of the triangle sum to 180°.

Example 2 — Why it is a right angle

Prove that the angle in a semicircle is 90°.



1. AB is a diameter, so it passes through the centre O.	diameter through O
2. A diameter is a straight line, so ∠AOB = 180°.	straight line at O
3. Angle at the centre = 2 × angle at the circumference.	∠APB = ½ × 180°

∠APB = 90°. The angle in a semicircle is a right angle — the special case of the angle-at-the-centre theorem when the arc is a semicircle. ■

See it hold, then prove it from the keystone

Let the learner drag P and watch the right angle refuse to change before stating the rule. Frame the proof explicitly as the special case of last lesson's theorem — a diameter is simply the arc whose central angle is 180° — so this lesson reinforces the keystone rather than adding a separate fact to memorise. Insist on the reason 'angle in a semicircle' with every use, and introduce the converse once the forward result is secure, as it is a common exam device for locating a diameter or a centre.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing

Connection	See the diameter give a straight angle at O and a right angle at P.	~9 min
Movement	Find the other angles of a triangle on a diameter, with reasons.	~11 min
Reflection	Circle Theorem Lab (diameter case): drag P, the 90° never moves.	~12 min
Creativity	Prove it in three lines from the keystone; meet the converse.	~9 min
Rest	A short pause on special cases doing extra work.	~2 min
Nutrition	Intellectual: a general rule aimed at its most special case.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Not spotting that AB is a diameter (so missing the 90°).
'Does the chord pass through the centre? If it is a diameter, the angle on it at the circumference is 90°.'

Giving 90° with no reason.
'Correct — and the reason is "angle in a semicircle". Say it every time.'

Forgetting the acute angles add to 90°, not 180°.
'The right angle already uses 90°. The other two share what is left of 180° — so they add to 90°.'

Using the theorem when the chord is not a diameter.
'Is that chord really a diameter through the centre? If not, this theorem does not apply — use the angle at the centre instead.'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the angle at the centre theorem (Lesson 2), that a diameter passes through the centre, and that angles on a straight line and in a triangle each sum to 180°. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner cannot start a triangle-on-a-diameter question, or gives 90° with no justification. The aim is a question that surfaces the diameter and the reason.

- "Is AB a diameter — does it go through the centre? Then what is the angle at P?"
- "What is your reason for the 90°?"

- “The right angle is 90°. How much is left for the other two angles together?”
- “If the angle at the circumference is 90°, what must the chord it stands on be?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why does the right angle stay 90° as P moves?

Because it always equals half the same 180° angle at the centre. The diameter fixes the central angle at 180° no matter where P is, so the inscribed angle is always 90°.

Is this really the same as last lesson?

Yes — it is the angle-at-the-centre theorem for the special arc that is a semicircle. Seeing it that way means one idea covers both, rather than two separate rules.

What is the converse and why is it useful?

If an inscribed angle is 90°, the chord it stands on is a diameter. That lets you locate a diameter (and hence the centre) from a right angle, or construct a right angle using a semicircle — a classic exam and construction tool.

Who was Thales?

Thales of Miletus (c. 600 BCE) is traditionally credited with this result, one of the oldest theorems in geometry; the fact is sometimes called Thales' theorem.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Supporting Adult Guidance v0.1.